EXPERIMENT 7: UNIFICATION AND RESOLUTION

AIM

To implement unification and resolution exercises using Prolog facts, compound terms, and rules.

PROCEDURE

1. Define the facts required for each exercise in Prolog.

2. Enter the query given for each exercise.

3. Execute the query and observe the variable bindings or truth value returned by Prolog.

CODE

1. Sentence Matcher

```
% Facts
sentence(the_cat_sits).
sentence(the_dog_runs).

% Query
?- sentence(X).
```

OUTPUT

```
X = the_cat_sits ;
X = the_dog_runs ;
false.
```

2. Compound Term Unification

```
% Facts
likes(john, food(pizza, spicy)).
likes(mary, food(pizza, sweet)).

% Query
?- likes(john, food(pizza, T)).
```

OUTPUT

```
T = spicy.
```

3. Conflict Resolution

```
% Facts
rule1(climate_change).
rule2(climate_change).

% Rule
conflict(X) :- rule1(X), rule2(X).

% Query
?- conflict(climate_change).
```

OUTPUT

```
true.
```

RESULT

Thus, the Prolog exercises for sentence matching, compound term unification, and conflict resolution were executed successfully.